Dylan Jacobs | djacobs2@swarthmore.edu | (503) 704-4583

Portfolio: https://dylan-jacobs.github.io/portfolio/ **GitHub**: github.com/dylan-jacobs **LinkedIn**: linkedin.com/in/dylan-t-jacobs/

Education

Swarthmore College, Philadelphia, PA Aug 2023—May 2027

Double-Major: BS in General Engineering (electrical engineering focus), BA in Applied Mathematics

Relevant coursework:

- Electronic circuit applications & analysis, Digital signal processing, Computer vision, Data structures & algorithms (C++)
- Ordinary & Partial differential equations (PDEs), Real Analysis, Tensor decompositions, Numerical PDEs
- Thermo-fluid mechanics, Electromagnetism, waves & optics, Statistical signal processing

Teaching Assistant for Electrical Circuit Analysis (ENGR 011) Aug 2025—present

Swarthmore Varsity Men's Soccer Aug 2023—Jun 2026

Overall GPA: 3.98/4.00

Research and work experience

Advanced Materials Intern & Research Assistant – US Naval Research Laboratory, Washington, DC May 2025-Aug 2025

- Measured thermal diffusivity and specific heat of thermoelectric semiconductors using laser flash analysis (LFA).
- Built Arduino-controlled LN_2 dispenser with closed-loop thermocouple monitoring, automating LFA experiments
- Created graphical-user-interface and data-logging interfacing between LN_2 dispenser and lab computer.
- Developed experiments to improve the lab's measurement procedure for IR-transparent and high-resistance samples.
- Poster presentations: Swarthmore *Sigma Xi* 2025 poster session.

Applied Mathematics Research Assistant – Swarthmore College Mathematics, Philadelphia, PA Jan 2024-present

- Utilizing computational fluid dynamics and numerical methods to research high-order accurate methods for partial differential equations (PDEs), plasma/kinetic models in MATLAB.
- Developing low-rank, structure-preserving integrator for Vlasov-Fokker-Planck plasma equation.
- Poster presentations: 2025 SIAM NNP conference, 2024 & 2025 *Sigma Xi* Swarthmore poster sessions.

Electrical Engineering Research Assistant – Swarthmore College Engineering, Philadelphia, PA Nov 2023—May 2024

- Researched electrical/aerospace technology behind wind-energy devices to develop oscillatory wind-energy harvester.
- Used Arduino, MATLAB & ViscousFlow to simulate air vortex-shedding & electrical induction power output.

Software Engineering Summer Intern - Oregon Health and Science University, Portland, OR Jun 2022—Aug 2022

- Used Kotlin to develop Android app that detects heart murmurs using audio data from a Bluetooth stethoscope.
- Presented machine-learning paper to the lab's reading group.

Projects

Convolutional Neural Network for Hand-Drawn Image Classification: 250 object classes, link May 2026

Analog Circuit and PCB Design: used Fusion360 to design and fabricate printed circuit boards (PCBs) Aug 2025—Dec 2025

- Combined MOSFETs, sensors, actuators, op-amps, comparators, audio amplifiers, and other integrated circuits
- Projects: audio equalizer, temp-controlled fan, light-optimizing solar panel, regulated DC power supply.

AI Python Stock Trading: trained neural network on technical indicators to trade stocks using Alpaca API, link Feb 2025

Generative Adversarial Network (GAN): Image-generating model trained on web-scraped art, link Apr 2022—Feb 2023

FireSale: Used Java, AWS backend to build Android app to reduce food waste and hunger, link 2020—2021

Technical & Professional Skills

Technical: SPICE, PCB design, CAD, Fusion360, Soldering, Embedded systems, PDE solvers, Numerical methods

Programming: Python, MATLAB, Arduino, Java, C++, HTML/JavaScript, LaTeX

Software: VSCode, Git, SolidWorks, AutoCAD, NumPy, TensorFlow, PyTorch

Languages: Spanish (Fluent), Global Seal of Biliteracy (2022), **Citizenship:** USA, Ireland

Honors and Scholarships

Tau Beta Pi Engineering Honors Society Mar 2026

Delaware Valley Engineers Undergraduate Scholarship, Delaware Valley Engineers Society Feb 2025

Donna Prentice Memorial Scholarship, American Society of Civil Engineers Feb 2024

National Merit Scholarship, National Merit Scholarship Corporation Apr 2023